Coupled Physical-Event Contracts for Robot Experience: Task-Graph Expansion and an Audited Component Study

Research development draft [1]

Abstract

Laboratory operations constrain physical effects, not merely command dispatch. Dependencies, persistent processes and original event references can make apparently timely actions violate the task. We develop independently scored event contracts and examine two components. An exact scheduling candidate matches exhaustive enumeration on 7,172 synthetic interval conditions; the preserved earlier planner incorrectly rejects 61 feasible conditions. A separate MuJoCo experiment executes 12 controlled trajectories across four task graphs, plus two separately reported sentinels. A reference-preserving graph controller satisfies all four registered contracts, compared with one for an unguarded graph controller and none for an independent-window ordering baseline. These fixed mechanical cases validate interfaces and expose failure modes. They do not establish learned-policy improvement, chemical outcomes, real-time agent execution or generalization. The proposed learning study must absorb verified recovery experience under matched budgets and independent evaluation.

1. Problem and motivation

Consider stopping addition within two seconds after an endpoint appears, or maintaining actual stirring for a prescribed duration while another operation shares the robot. The deadline applies to an effective event such as stopped flow or motion. A prompt command can still produce a late physical effect; restarting a process can create a timely-looking result while abandoning the original experiment. We propose studying experience generation, verification and absorption under these original physical contracts.

The benchmark's C5, C6 and C13 definitions remain unchanged. C5 requests a two-second endpoint-to-stop response; C6 specifies 120 ± 10 seconds of actual stirring; C13 includes a transient A cue and randomized A-to-B waiting in a no-clock track. The current twin-rotor fixture is separately registered and does not certify any of these chemical tasks. Preserved A-r7 robot timing evidence also remains separate: one fixed instance and three earlier trials.

Positioning. Physical-time representations are already studied by Chronos (Chronos, 2026), while remaining-time control, punctuality, recovery and coordination are considered by Time Control (2026). Agent context and robot self-improvement are studied by GPT-Policy and RoboRSI (GPT-Policy, 2026; RoboRSI, 2026). Thus adding time to DP or assembling an agent training pipeline is insufficient as a novelty claim. Our proposed emphasis is the interaction of independently measured event contracts, immutable original references, concurrent process preservation and verified experience absorption. The component evidence below supports infrastructure development, not the full proposed contribution.

2. Physical-event formulation

Let a task contain an event dependency graph G, contracts Φ, observation rules O and resource capacities R. Observation capture and delivery, command dispatch and physical effect have distinct timestamps. The actor uses only permitted delivered observations, public task information and past legal history. An independent scorer uses raw physical measurements, which are not passed back as privileged actor inputs.

An event occurrence is a conservative interval $I_e = [\ell_e, u_e]$ on one identified physical clock. Confirmation time is retained separately. For an event-to-event target D and tolerance ϵ, score

$$I_{\text{error}} = [\ell_b - u_a - D,\ u_b - \ell_a - D].$$

TRUE requires containment in $[-\epsilon, \epsilon]$; disjointness is FALSE; overlap without containment is UNKNOWN. Missing and left-censored events cannot count as success. Formal joint success conjoins timing, required physical predicates, dependency, original-reference and safety conditions. Strict

[1] Component report; shared-policy and chemical evaluation remain open. Correspondence to: Author details pending.

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sampled maintenance is reported explicitly; it does not establish continuity between physical samples.

For the proposed general system, the reference ledger retains already established original anchors. Recovery cannot relabel a reference or reset a process and count it as the same task. An infeasible original contract remains failed or safely stopped; a restarted process is a new task instance. Cold reconstruction does not establish continuous process time without an independent continuous clock and evolution.

3. Joint scheduling component

For a public cue c, desired contact after D, contact-to-release gap H, known FIFO delay δ and mechanical latency boxes $a \in [a_L, a_U]$, $b \in [b_L, b_U]$, feasible grid dispatches satisfy

$$\begin{aligned} d_A - c &\in [D - \epsilon - \delta - a_L,\ D + \epsilon - \delta - a_U], \\ d_B - c &\in [D + H - \epsilon - \delta - b_L,\ D + H + \epsilon - \delta - b_U], \\ d_B - d_A &\in [H - \epsilon - b_L + a_U,\ H + \epsilon - b_U + a_L]. \end{aligned}$$

The preserved skill also imposes a minimum stage gap and an earliest contact dispatch. The same constant FIFO delay for both events cancels from the gap but not from the absolute windows. Two separate absolute-window passes permit opposing errors and therefore do not imply the gap contract.

The candidate uses exact rational constraints. For each feasible contact grid tick, intersect the release grid bounds with the gap and minimum-skill bounds. The objective minimizes squared distance to midpoint dispatches, with deterministic grid tie-breaking. For fixed contact, the convex release term attains its discrete minimum at a clamped floor/ceiling of the ideal release; checking these points suffices. Independently enumerated all-pair constraints and a third integer implementation check the returned feasibility and choice. This is classical finite scheduling, not a new general safety theorem. Soundness applies only to the supplied box, resource and skill assumptions; an empirical calibration envelope is not true uncertainty support.

The source and roster were frozen before the full matrix. It contains 6,912 factorial conditions, 256 fixed-seed synthetic draws and four targeted boundaries. Grid periods, latency widths, delay, tolerances and skill minimums vary. Every exception and refusal remains in the 7,172-condition denominator. Synthetic draws are not physical random seeds or robot episodes.

The candidate matches complete enumeration in every condition, including 4,566 infeasible cases. Independent enumeration checks 299,040 grid pairs. The old joint planner falsely refuses 61 feasible cases, because it rejects a negative independent-midpoint release hold before its joint search. Its scheduled outputs have no erroneous robust acceptance in this matrix. The independent midpoint schedules also have 994 cases in which both absolute event windows pass while their gap fails. The candidate remains separate from the preserved native A source; the latter is not silently fixed.

Table 1. Synthetic interval conditions. Denominator 7,172 in every row; 2,606 conditions are mathematically feasible. No physical episode credit.

Method	Robust joint	Exceptions
Nominal prior	409	975
Public delay	788	981
Independent midpoint	2375	1198
Old joint	2545	1198
Exact candidate	2606	0

4. Task-graph physics component

A preserved contact-free twin-rotor MJCF is integrated by MuJoCo at 10 ms. Four registered graphs are parallel, A-before-B, B-before-A and A-while-B-active. A shared serial command bus and a global relay have measured effective control schedules. All three controllers receive the same delivered state, public graph, service model and fixed calibration set. Private true dynamics, reference truth and scorer labels are withheld. Dynamics and fault setting are fixed inside the calibration set; this is a development component with no multi-seed or learned task-space generalization claim.

The ordering baseline retains an A-before-B command form and checks independent endpoint windows. The graph scheduler searches graph-constrained command forms and can use the global relay. The guarded variant additionally protects predicted original reference acquisition. These are hand-written controllers, not learned policies or a reproduction of an earlier history agent.

Each of 12 main trajectories and two sentinels integrates one complete ten-second epoch: 14,014 controlled state samples. One independently reconstructed full-power trajectory supplies counterfactual benchmark reference crossing intervals (1,001 diagnostic samples); 100 preflight steps are counted separately. This fixture compares actual crossings with that fixed counterfactual bracket. It does not validate a ledger recovery for an externally observed, already occurred chemical event. No task-period state/time write or reset occurs. Stops require a previously moving rotor followed by at least 200 ms of low-speed samples: 21 points, with onset brackets distinct from confirmation. B preservation conjoins sampled torque and motion predicates until its required original bound. The independent audit re-integrates all controls and recomputes all contracts.

The feasible sentinel satisfies its original contract. The dependency-cycle sentinel correctly produces no commands, integrates its full trace, and remains FALSE in physical con-

Table 2. Main mechanical contracts: four instances per controller. The two sentinels are excluded from this table and reported separately.

Controller	TRUE	FALSE	UNKNOWN
Independent windows	0	4	0
Graph constraints	1	3	0
Graph + reference guard	4	0	0

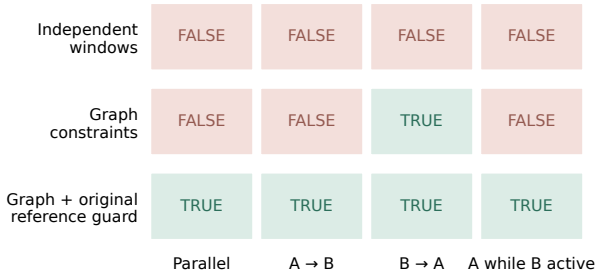


Figure 1. Independent original-contract verdicts for the four registered graphs. Fixed mechanical cases; hand-written rules; no new weights.

tract scoring; correct refusal is separately recorded. All twelve main cases remain in their original denominator; both sentinels are reported separately. The controlled dispatch/delivery delays advance native time, but host planning work pauses this fixture. Consequently these results do not establish physical continuity during real model waits or wall-clock timing guarantees.

5. Proposed learning study and open gates

The agent should compile tasks, choose data, diagnose failures and propose recoveries. A fast executor preserves physical processes; the independent verifier determines data admission. Verified successes, failures and valid recoveries retain original anchors in replay. The initial learning study compares shared DP inputs under one backbone: no added time, legal public time, capture/execution timestamps and past history with reset/shuffle controls.

Teacher relabeling on student-visited states must match teacher identity, labels, optimization, native steps and compute to offline imitation. If the update is supervised aggregation, it is DAgger-style (Ross et al., 2011); a distinct on-policy distillation objective must be specified before claiming OPD. Offline and online RL are conditional ablations, rather than mandatory stages. Context reset tests parameter absorption; frozen weights with retained history test context adaptation. Evaluation splits full instances, not frames, and separates state, time, graph and process-drift holdouts.

The frozen A-r7 extension plans four calibration and 16 tests, varying bottle mass/inertia, actual position-target FIFO and joint contact/release contracts. It has no admitted new test result in this component report. After initial resource refusal, its first calibration starts when one GPU satisfies the frozen 24 GiB admission. We neither lower that bound nor preempt foreign jobs. The four-calibration empirical envelope must pass independent raw audit before test use.

Remaining scientific gates include calibrated flow/motor/thermal endpoints, continuous process evolution during real planner waits, independently clocked hardware, actual shared policy weights and matched-budget held-out improvements. The present component tests expose infrastructure errors and make the proposed study reviewable; they do not complete the full method-paper claim.

References

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